In [37]:
```
import pandas as pd
df=pd.read_csv("Amazon.csv")
df.head(5)
```

Out[37]:

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName	Catego
0	ORD0000001	2023-01-31	CUST001504	Vihaan Sharma	P00014	Drone Mini	Boo
1	ORD0000002	2023-12-30	CUST000178	Pooja Kumar	P00040	Microphone	Home Kitch
2	ORD0000003	2022-05-10	CUST047516	Sneha Singh	P00044	Power Bank 20000mAh	Clothi
3	ORD0000004	2023-07-18	CUST030059	Vihaan Reddy	P00041	Webcam Full HD	Home Kitch
4	ORD0000005	2023-02-04	CUST048677	Aditya Kapoor	P00029	T-Shirt	Clothi

In [38]:
```
df.tail(5)
```

Out[38]:

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName	C
99995	ORD0099996	2023-03-07	CUST001356	Karan Joshi	P00047	Memory Card 128GB	El
99996	ORD0099997	2021-11-24	CUST031254	Sunita Kapoor	P00046	Car Charger	C
99997	ORD0099998	2023-04-29	CUST012579	Aman Gupta	P00030	Dress Shirt	C
99998	ORD0099999	2021-11-01	CUST026243	Simran Gupta	P00046	Car Charger	C
99999	ORD0100000	2021-12-04	CUST029492	Sunita Reddy	P00019	LED Desk Lamp	

In [39]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 20 columns):
 #   Column         Non-Null Count   Dtype  
---  ------         --------------   -----  
 0   OrderID        100000 non-null  object 
 1   OrderDate      100000 non-null  object 
 2   CustomerID     100000 non-null  object 
 3   CustomerName   100000 non-null  object 
 4   ProductID      100000 non-null  object 
 5   ProductName    100000 non-null  object 
 6   Category       100000 non-null  object 
 7   Brand          100000 non-null  object 
 8   Quantity       100000 non-null  int64  
 9   UnitPrice      100000 non-null  float64
 10  Discount       100000 non-null  float64
 11  Tax            100000 non-null  float64
 12  ShippingCost   100000 non-null  float64
 13  TotalAmount    100000 non-null  float64
 14  PaymentMethod  100000 non-null  object 
 15  OrderStatus    100000 non-null  object 
 16  City           100000 non-null  object 
 17  State          100000 non-null  object 
 18  Country        100000 non-null  object 
 19  SellerID       100000 non-null  object 
dtypes: float64(5), int64(1), object(14)
memory usage: 15.3+ MB
```

In [40]:
```
print("The number of rows",df.shape[0])
```

```
The number of rows 100000
```

In [41]:
```
print("The number of columns",df.shape[1])
```

```
The number of columns 20
```

In [42]:
```
print("The names of columns",df.columns)
```

```
The names of columns Index(['OrderID', 'OrderDate', 'CustomerID', 'CustomerName', 'P
roductID',
       'ProductName', 'Category', 'Brand', 'Quantity', 'UnitPrice', 'Discount',
       'Tax', 'ShippingCost', 'TotalAmount', 'PaymentMethod', 'OrderStatus',
       'City', 'State', 'Country', 'SellerID'],
      dtype='object')
```

In [43]:
```
df.isnull()
```

Out[43]:

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName	Categ
0	False	False	False	False	False	False	Fa
1	False	False	False	False	False	False	Fa
2	False	False	False	False	False	False	Fa
3	False	False	False	False	False	False	Fa
4	False	False	False	False	False	False	Fa
...	...	...	...	...	...	...	
99995	False	False	False	False	False	False	Fa
99996	False	False	False	False	False	False	Fa
99997	False	False	False	False	False	False	Fa
99998	False	False	False	False	False	False	Fa
99999	False	False	False	False	False	False	Fa

100000 rows × 20 columns

In [44]:

```
df.dropna()
```

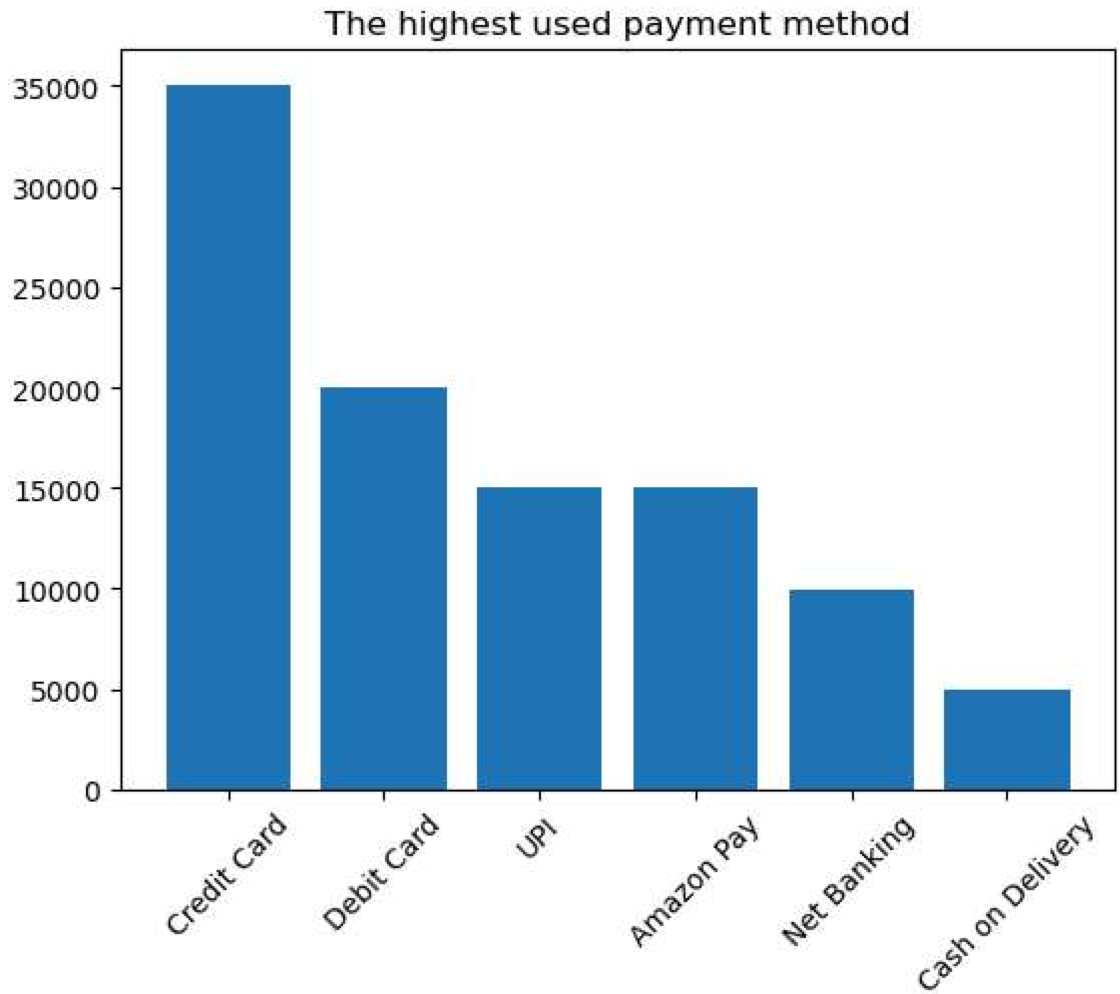
Out[44]:

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName
0	ORD0000001	2023-01-31	CUST001504	Vihaan Sharma	P00014	Drone Mini
1	ORD0000002	2023-12-30	CUST000178	Pooja Kumar	P00040	Microphone
2	ORD0000003	2022-05-10	CUST047516	Sneha Singh	P00044	Power Bank 20000mAh
3	ORD0000004	2023-07-18	CUST030059	Vihaan Reddy	P00041	Webcam Full HD
4	ORD0000005	2023-02-04	CUST048677	Aditya Kapoor	P00029	T-Shirt
...	...	...	...	...	...	...
99995	ORD0099996	2023-03-07	CUST001356	Karan Joshi	P00047	Memory Card 128GB
99996	ORD0099997	2021-11-24	CUST031254	Sunita Kapoor	P00046	Car Charger
99997	ORD0099998	2023-04-29	CUST012579	Aman Gupta	P00030	Dress Shirt
99998	ORD0099999	2021-11-01	CUST026243	Simran Gupta	P00046	Car Charger
99999	ORD0100000	2021-12-04	CUST029492	Sunita Reddy	P00019	LED Desk Lamp

100000 rows × 20 columns

In [45]:

```
import matplotlib.pyplot as plt
c=df["PaymentMethod"].value_counts()
plt.title("The highest used payment method")
plt.bar(c.index,c.values)
plt.xticks(rotation=45)
plt.show()
```



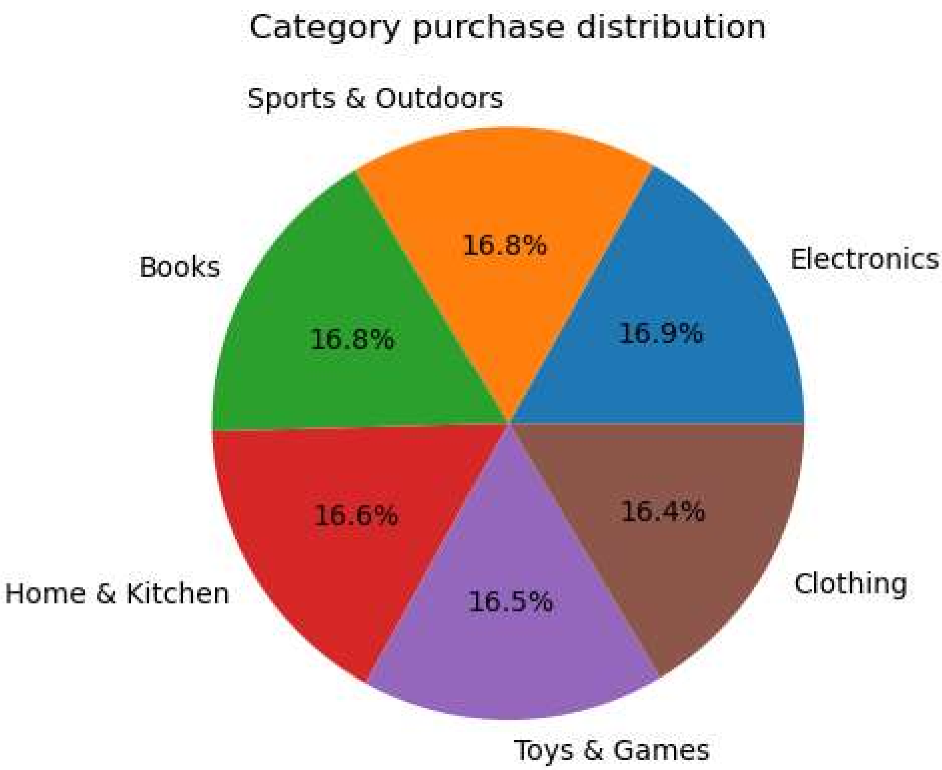
In [48]:
```
df["Final"]=df["UnitPrice"]*df["Quantity"]*(1-df["Discount"])+df["Tax"]+df["Shippin
df.head()
```

Out[48]:

	OrderID	OrderDate	CustomerID	CustomerName	ProductID	ProductName	Catego
0	ORD0000001	2023-01-31	CUST001504	Vihaan Sharma	P00014	Drone Mini	Boo
1	ORD0000002	2023-12-30	CUST000178	Pooja Kumar	P00040	Microphone	Home Kitch
2	ORD0000003	2022-05-10	CUST047516	Sneha Singh	P00044	Power Bank 20000mAh	Clothi
3	ORD0000004	2023-07-18	CUST030059	Vihaan Reddy	P00041	Webcam Full HD	Home Kitch
4	ORD0000005	2023-02-04	CUST048677	Aditya Kapoor	P00029	T-Shirt	Clothi

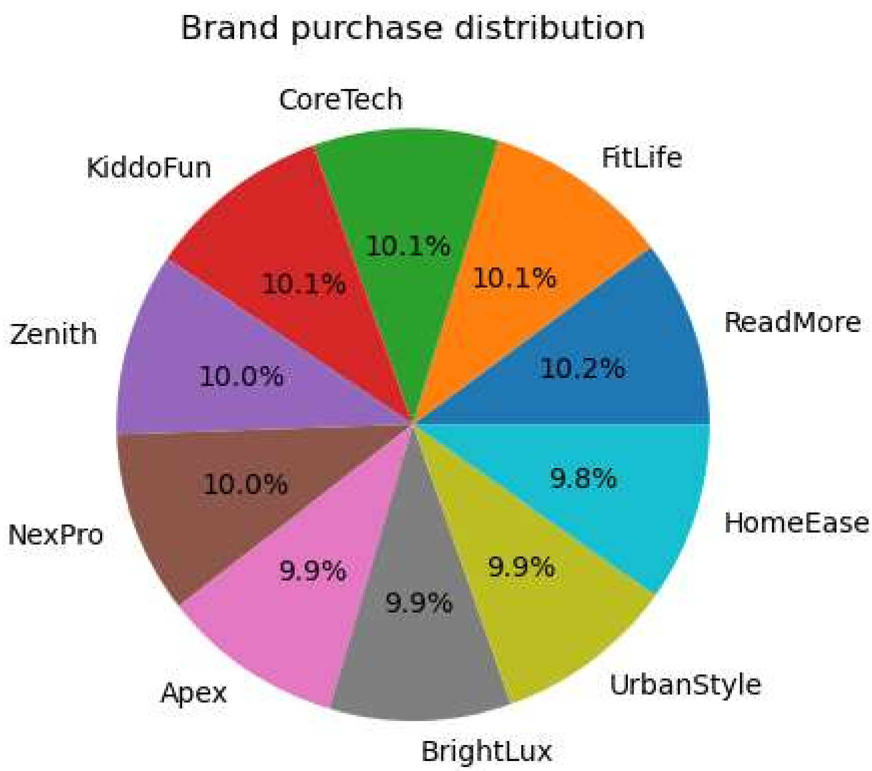
5 rows × 21 columns

In [35]:
```
d=df["Category"].value_counts()
plt.title("Category purchase distribution")
plt.pie(d.values,labels=d.index,autopct="%1.1f%%")
plt.show()
print("This helps us to identify which category is preferred by customers")
```




```
This helps us to identify which category is preferred by customers
```

In [36]:
```
f=df["Brand"].value_counts()
plt.title("Brand purchase distribution")
plt.pie(f.values,labels=f.index,autopct="%1.1f%%")
plt.show()
print("This helps us to get the useful information like which brand is purchased th
```

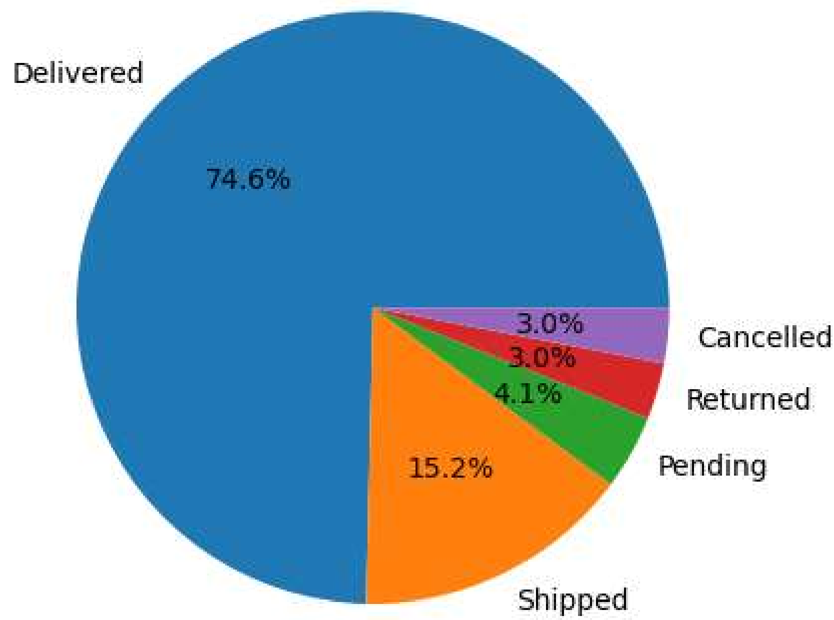


This helps us to get the useful information like which brand is purchased the most

In [52]:
```
print("The product that is purchased most is ", df["ProductName"].value_counts().id
```

The product that is purchased most is LED Desk Lamp

In [54]:
```
g=df["OrderStatus"].value_counts()
plt.pie(g.values,labels=g.index,autopct="%1.1f%%")
plt.show()
```

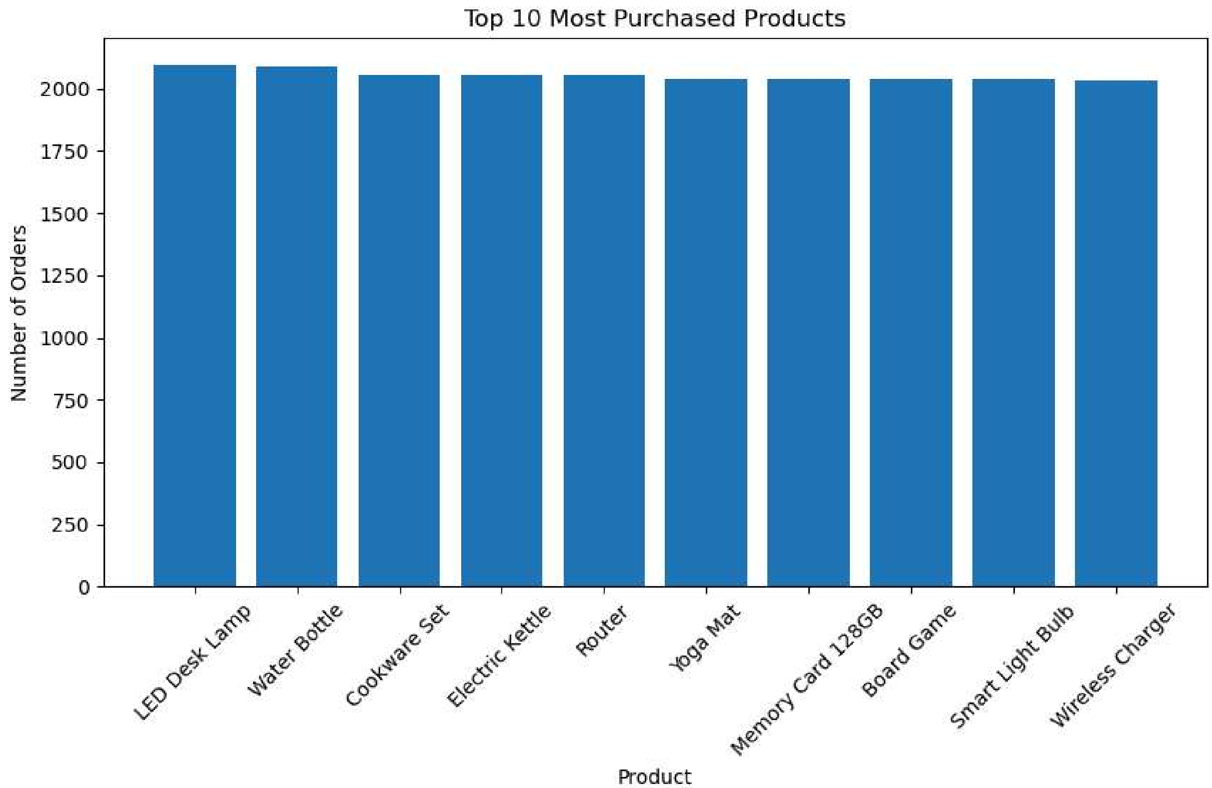


In [55]:

```
top_products = df["ProductName"].value_counts().head(10)

plt.figure(figsize=(10,5))
plt.bar(top_products.index, top_products.values)
plt.xticks(rotation=45)
plt.title("Top 10 Most Purchased Products")
plt.xlabel("Product")
plt.ylabel("Number of Orders")
plt.show()

print("Insight: These are the products purchased most frequently.")
```

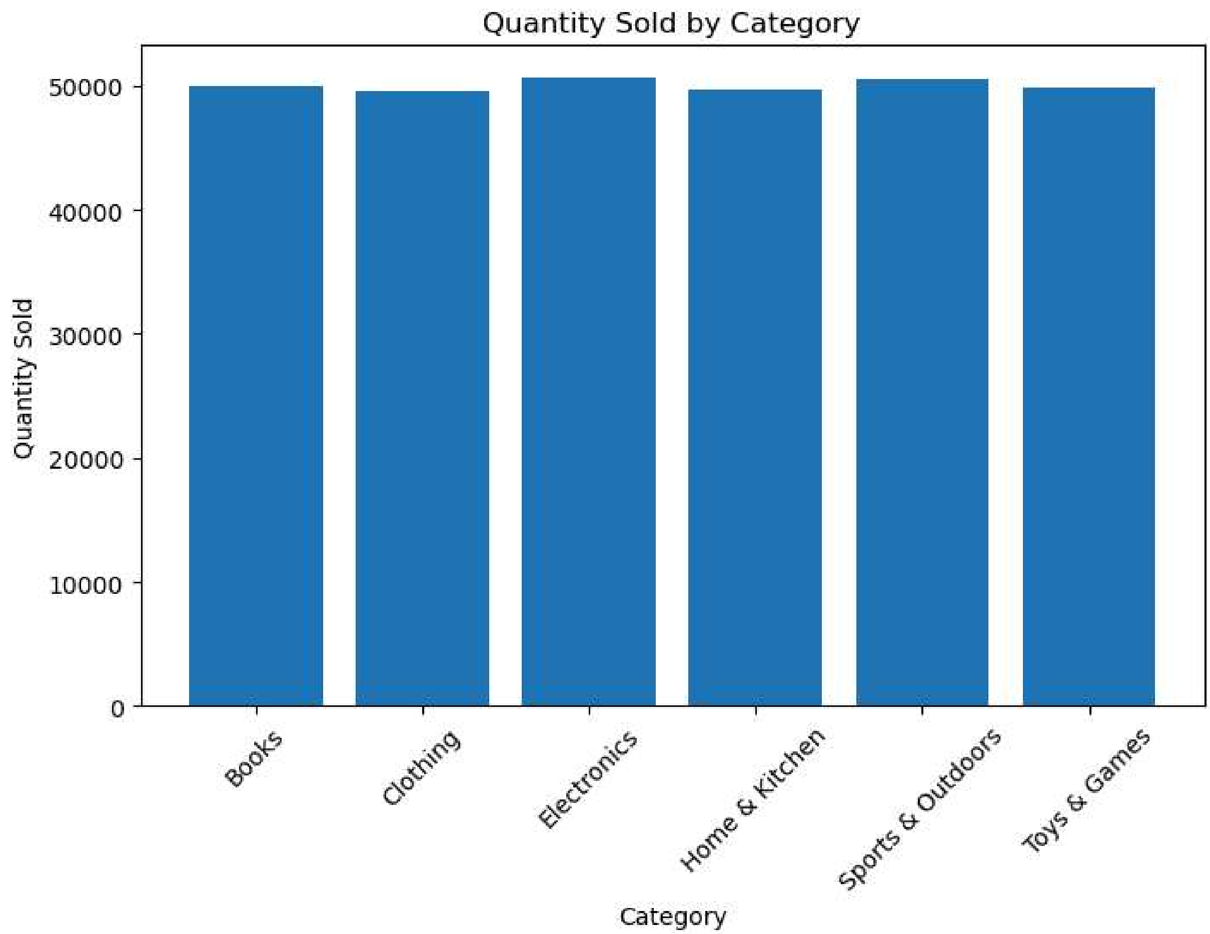


Insight: These are the products purchased most frequently.

In [58]:
```
quantity = df.groupby("Category")["Quantity"].sum()

plt.figure(figsize=(8,5))
plt.bar(quantity.index, quantity.values)
plt.xticks(rotation=45)
plt.title("Quantity Sold by Category")
plt.xlabel("Category")
plt.ylabel("Quantity Sold")
plt.show()

print("This graph shows which categories sell the highest number of items.")
```



This graph shows which categories sell the highest number of items.

```
In [ ]:
```